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Status and influencing factors of academic burnout among undergraduate nursing students based on random forest modeling: a cross-sectional study

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Abstract

Background Academic burnout is a widespread issue among nursing students, significantly impeding the development of high-quality nursing professionals, particularly among undergraduate nursing students. This study aims to investigate the status of academic burnout in this population and to analyze the factors contributing to it.

Methods This study utilized convenience sampling to conduct a cross-sectional survey involving 442 undergraduate nursing students. Data were collected using the Socio-demographic Information Questionnaire, the Academic Burnout Scale (ABS), the Time Management Disposition Scale (TMD), and the Self-Regulatory Fatigue Scale (SRF-S). Random Forest and LASSO regression were employed to rank the importance of various factors and identify key variables. Subsequently, multivariate linear regression was conducted to examine the determinants of academic burnout among undergraduate nursing students.

Results The average academic burnout score for undergraduate nursing students was 55.29 ± 6.78 , with a prevalence of 57.5%. Results from Random Forest and LASSO regression analyses identified the eight most significant predictive variables, ranked in order of importance: Self-Regulatory Fatigue, study time per day, time management disposition, absence of study interest in nursing-related fields, academic performance, being an only child, future professional prospects and excessive course difficulty. Furthermore, multivariate linear regression analysis revealed that Self-Regulatory Fatigue, study time per day, time management disposition, academic performance, future professional prospects, and excessive course difficulty are the primary factors influencing academic burnout among undergraduate nursing students.

Conclusions Undergraduate nursing students demonstrate above-average levels of academic burnout, highlighting the need for significant attention to this issue. Nursing educators should prioritize academic burnout prevention and design targeted interventions based on the identified key influencing factors to mitigate its prevalence.

Keywords Undergraduate nursing students, Academic burnout, Self-regulatory fatigue, Random forest, LASSO regression, Influencing factors

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Background

Burnout was first introduced by American psychologist Herbert Freudenberger in the 1970s to describe a state of exhaustion resulting from prolonged working hours, excessive workloads, and low job satisfaction among helping professionals, such as teachers, nurses, and social workers [1]. Subsequent research has expanded the concept of burnout to include academic settings, indicating that it also affects student populations [2], a phenomenon referred to as academic burnout [3, 4].

Academic burnout is characterized by a gradual decline in guilt, emotional exhaustion, and enthusiasm for learning, as defined by [5]. This phenomenon typically arises from excessive academic demands and persistent academic stress, which can adversely affect the learning abilities and academic performance of nursing students [6]. Consequently, students may experience a diminishing interest in their studies. The average detection rate of academic burnout among Chinese nursing students is 44.83%, with undergraduate nursing students exhibiting an even higher prevalence of 51.3% [7]. These statistics are concerning and warrant further investigation.

The consequences of academic burnout can encompass a range of psychological and physical issues, including weakness and insomnia; emotional challenges such as anxiety and depression; attitudinal difficulties characterized by hostility, indifference, and mistrust; and behavioral problems like aggression and nervousness [8]. Students experiencing academic burnout often face negative emotions, including psychological withdrawal and academic frustration, driven by a fear of underachievement [9]. These adverse feelings affect their attitudes toward learning and academic performance, further exacerbating burnout. Moreover, academic burnout is recognized as a significant predictor of mental health [10] and a factor that hinders job readiness and contributes to brain drain [11]. Consequently, it is essential to identify the risk and protective factors associated with academic burnout among undergraduate nursing students.

Time management disposition reflects individual personality differences related to temporal aspects, and are considered a personal trait. Positive personality traits can diminish the likelihood of negative behaviors [12] and may serve as significant protective factors against academic burnout. Self-Regulatory Fatigue [13] refers to a state of impaired executive function resulting from the depletion of an individual's psychological resources for self-control, manifesting as internal resource exhaustion. According to the Job Demands-Resources (JD-R) model adapted for higher education students [14, 15], it is posited that specific factors influence burnout, categorized into job demands and job resources. Job demands encompass the organizational and psychosocial aspects of work that require cognitive and emotional effort, often

associated with social or psychological costs. Conversely, job resources include the physical, psychosocial, and organizational elements that facilitate task achievement, professional growth, and personal development, helping to mitigate job demands and their associated psychological or physiological costs. Our analysis, grounded in academic characteristics, suggests that excessive job demands, such as overloaded study hours, pressure to perform academically, excessive course difficulty, and Self-Regulatory Fatigue combined with insufficient job resources, like time management disposition and positive future professional prospects generate academic stress, thereby contributing to academic burnout. Consequently, this study investigates the academic characteristics of undergraduate nursing students through the lens of the Job Demands-Resources (JD-R) model and explores the mechanisms by which these characteristics influence academic burnout. This research has significant implications for promoting academic development and enhancing the mental health of undergraduate nursing students.

Existing literature indicates that scholars have investigated the prevalence of academic burnout multiple times, yielding a wealth of research findings. Most existing research has primarily concentrated on junior high and high school students, as well as college students [16, 17]. However, the unique characteristics of the nursing profession limit the generalization of these findings to undergraduate nursing students. Furthermore, most studies have employed regression analysis to identify factors influencing academic burnout, yet few have ranked the importance of these factors or identified the primary influences.

Random Forest, a comprehensive machine learning algorithm introduced by Breiman in 2001, is well-suited for classification and regression tasks. It effectively handles missing values, boasts a high accuracy rate, and allows for the assessment of the importance of various factors, has been widely used in the medical field in recent years [18, 19]. On the other hand, LASSO regression is a variable selection method known for its stability, which simplifies models by introducing a penalty term during estimation [20]. Therefore, this study aims to evaluate the importance ranking of various factors by constructing a random forest model. In conjunction with LASSO regression, it seeks to identify the primary contributors to academic burnout among undergraduate nursing students.

Undergraduate nursing students represent a vital future workforce for clinical nursing. It is essential to prevent these students from entering the profession experiencing burnout, as this condition can adversely affect not only their well-being but also the quality of patient care and the efficacy of healthcare organizations. Accordingly, understanding the status of academic burnout

among undergraduate nursing students and identifying its primary influencing factors can serve as a foundation for developing targeted interventions. These strategies aim to effectively alleviate academic burnout and reduce turnover rates among nursing professionals.

The objectives of this study are: (1) assess the levels of academic burnout among undergraduate nursing students; (2) rank the relative importance of the various factors influencing academic burnout among undergraduate nursing students; and (3) identify and determine the primary factors contributing to academic burnout among undergraduate nursing students.

Methods

Design and collection

A cross-sectional research design was employed, utilizing a convenience sampling method. Electronic questionnaires were distributed via Questionnaire Star (<http://www.wjx.cn>). Permission was obtained from school administrators, and volunteers—including counselors, full-time faculty members, and student representatives were recruited prior to the survey. After completing standardized online training, these volunteers distributed the questionnaires to undergraduate nursing students after school hours. The first page of the questionnaire included a guide that provided informed consent, an overview of the questionnaire, the survey's purpose, and instructions for completing the questionnaire. Participants were considered to have consented to the study by beginning to fill out the questionnaire after reading the informed consent section. To ensure the integrity of data, each participant was allowed to complete only one questionnaire, and all questions had to be answered before submission. The survey process ensured informed consent, voluntary participation, and anonymous submission. A total of 480 questionnaires were distributed, resulting in 442 valid responses, yielding an effective response rate of 92.08%.

Participants

This study was conducted from June to November 2024 across three medical schools offering undergraduate nursing programs in Sichuan Province, Jiangsu Province, and the Inner Mongolia Autonomous Region, with a total of 442 undergraduate nursing students participating. The inclusion criteria of participants were: (a) enrolled full-time four-year undergraduate students; (b) providing informed consent for voluntary participation in the study. Exclusion criteria: (a) suspension or withdrawal from school; (b) the presence of major physical illnesses or mental disorders during the investigation period.

Instruments

Social-demographic information questionnaire

Drawing on a literature review and expert consultation, members of our group worked together to design socio-demographic information on undergraduate nursing students, including gender, grade, birthplace, whether they are an only child, the main reason for choosing nursing, student leader, academic performance, barriers to learning, and study time per day, professional recognition and future professional prospects.

Academic burnout scale (ABS)

The Academic Burnout scale (ABS) was developed by [21] which is a national scale used to measure the level of academic burnout among nursing students. The ABS comprises three dimensions: dejection (8 items), improper behavior (6 items), and reduced personal accomplishment (6 items). A 5-point Likert scale is employed, ranging from 1 (not at all compliant) to 5 (fully compliant). A score above 60 indicates the presence of academic burnout, whereas a score below 40 signifies the absence of academic burnout. Scores that fall between 40 and 60 represent an uncertain intermediate state. Previous studies have demonstrated that the ABS possesses strong reliability and validity. In this study, the Cronbach's reliability of ABS was 0.822.

Time management disposition scale (TMD)

Time Management Disposition Scale (TMD) was developed by [12] to reflect the cultural context and specific national conditions of China. The TMD comprises 44 items distributed across three dimensions: Sense of Time Value (10 items), Sense of Time Control (24 items), and Sense of Time Efficacy (10 items). Participants respond using a 5-point Likert scale ranging from 1 (not at all compliant) to 5 (completely compliant), resulting in total scores that can range from 44 to 220. A higher score indicates better time management skills. A mean score above 3 reflects a tendency toward moderate or better time management, while a mean score below 3 indicates below-moderate tendencies. Previous studies have established the TMD's good reliability and validity. In this study, the Cronbach's reliability of TMD was 0.803.

Self-regulatory fatigue scale (SRF-S)

In this study, we utilized the Chinese version of the Self-Regulatory Fatigue Scale (SRF-S) translated and revised by [22]. The SRF-S comprises three dimensions: cognitive control, emotional control, and behavioral control, totaling 16 items. Responses are measured using a 5-point Likert scale, with reverse-scored items rated from 5 (strongly disagree) to 1 (strongly agree). Higher scores on the scale indicate greater levels of Self-Regulatory Fatigue among nurses. Mean scores above 3 suggest higher than

moderate levels of Self-Regulatory Fatigue, while scores below 3 indicate lower than moderate levels. Previous studies have demonstrated that the SRF-S exhibits good reliability and validity. In this study, the Cronbach's reliability of SRF-S was 0.918.

Data analysis

Random Forest is a machine learning algorithm that operates on the principles of classification trees. It utilizes bootstrapping to randomly sample *n* training samples from the original dataset, designating the unsampled samples as out-of-bag (OOB) data for testing. OOB data is essential for evaluating the model's generalization capabilities and for parameter tuning, allowing for a more accurate assessment of Random Forest performance and enhancing both predictive precision and reliability. The *n* training samples are used to construct *n* classification trees that together form the Random Forest, with the final classification of test samples determined by majority voting among the classification tree results. In this study, 70% of the dataset is allocated as the training set, while the remaining 30% serves as the test set. This division ensures the accuracy of the trained model, with its discriminative capability assessed through the area under the receiver operating characteristic curve (AUC). A significance level of $\alpha=0.05$ is used for the construction of the Random Forest model. This model evaluates the importance of predictor variables using the increase in mean square error (%IncMSE). Specifically, each predictor variable is assigned a random value, and if a predictor is crucial, the model's prediction error increases when its value is randomly replaced. A higher %IncMSE indicates a greater impact of that variable on the model [23]. In this study, a Random Forest model was developed with academic burnout as the dependent variable, and it incorporated 18 characteristic variables, including demographic and sociological factors, Self-Regulatory Fatigue, and time management disposition. The weight values of these variables were calculated and ranked to ascertain their relative importance. This methodology facilitates the retention of significant variables while eliminating those with minimal or insignificant contributions, thereby enhancing the predictive performance of the model.

LASSO regression is a variable selection method known for its high model stability. By incorporating a

penalty term in the model estimation, it continuously compresses the coefficients. The trajectory of regression coefficients is represented by distinct curves, each illustrating the path of the coefficients for individual independent variables. The penalty coefficient is defined as the sum of the absolute values of the regression coefficients, leading to the compression of coefficients associated with characteristic variables that exhibit weak correlations with academic burnout to zero, thereby facilitating variable selection. The optimal lambda(λ), which corresponds to the smallest deviation in the cross-validation curves, was selected as the value at which the LASSO regression model achieved the best fit [20].

Statistical Methods. Random Forest modeling is effective in ranking the importance of factors influencing academic burnout. Its integration with LASSO regression aids in identifying the primary influences on academic burnout, while subsequent application of multiple linear regression enhances the validity of the findings. The Random Forest model was developed using the "Random Forest" package in R Studio, while LASSO regression was implemented through the "glmnet" package in R Studio. Descriptive statistics were conducted using SPSS version 26.0 to present the Mean $\pm$ SD of undergraduate nursing students' academic burnout and its dimensions. The Kolmogorov-Smirnov test was employed to verify data normality, upon testing, The data in this study were found to be approximately normally distributed; therefore, independent samples t-tests and one-way analysis of variance (ANOVA) were employed to compare academic burnout scores among nursing students with differing characteristics. Multifactorial analysis was conducted using the multiple linear regression method. Differences were deemed statistically significant at $P<0.05$ (two-sided).

Results
Academic burnout

Undergraduate nursing students reported an average total score of (55.29 $\pm$ 6.78) on the academic burnout, with 254 students (57.5%) experiencing academic burnout. Among the three dimensions of academic burnout, the highest score was observed in reduced personal accomplishment (2.84 $\pm$ 0.96), followed by improper behavior (2.81 $\pm$ 0.88), while the lowest score was recorded in dejection (2.66 $\pm$ 0.95) (see Table 1).

General information and academic burnout scores

The results indicated that scores on the academic burnout varied among undergraduate nursing students based on several factors, including gender, grade, birthplace, whether they were only child, study time per day, barriers to learning, academic performance, parents working in medical professions, professional recognition and Future

Table 1 Academic burnout scores (N=442)

Variables	Mean score	Score
TAB	2.76 $\pm$ 0.82	55.29 $\pm$ 6.78
DE	2.66 $\pm$ 0.95	21.31 $\pm$ 4.36
IB	2.81 $\pm$ 0.88	16.89 $\pm$ 3.25
RPA	2.84 $\pm$ 0.96	17.10 $\pm$ 3.27

Abbreviations: DE, dejection; IB, improper behavior; RPA, reduced personal accomplishment; TAB, total academic burnout

professional prospects, time management disposition, and Self-Regulatory Fatigue ($P < 0.05$). (Table 2)

Screening for factors influencing academic burnout Variable importance ordering

Using academic burnout as the dependent variable, we incorporated the 20 statistically significant variables identified in the univariate analysis as independent variables into the Random Forest model. To further assess the importance of each influencing factor, we coded these variables as V1 through V20. We randomly selected 70% of the organized data as the training set and 30% as the test set. Through parameter tuning, we determined that the out-of-bag (OOB) error rate was minimized (0.0405) when the number of decision trees (ntree) was set to 500, and the number of features selected (mtry) was set to 3. Additionally, results demonstrated that the Random Forest model achieves stability when the number of decision trees exceeds 500 [24]. Under these parameters, the model attained an area under the receiver operating characteristic curve (AUC) of 0.852 (95% CI: 0.715–0.982), indicating strong fitting capability. The %Inc MSE metric indicates the average reduction in precision, where a larger %Inc MSE signifies greater importance of the variable among the influencing factors [25]. The results indicate that among undergraduate nursing students, the order of importance from highest to lowest is as follows: Self-Regulatory Fatigue, study time per day, time management disposition, lack of Interest in Nursing-Related Fields, academic performance, Only Child, Future professional Prospects, excessive course difficulty, level of recognition, birthplace, Grade, Insufficient Learning Resources, and Excessive Social Pressure. See Fig. 1.

Variable selection

In accordance with the variable importance ranking results, the LASSO analysis was conducted using the glmnet in R Studio on 18 variables that demonstrated statistically significant differences in the Univariate analysis. Determine the optimal parameter λ by setting up 10-fold cross-validation, as illustrated in Fig. 2. The vertical dashed line on the left side of the figure indicates $\lambda_{\min}$, while the vertical dashed line on the right side represents λ_{1se} . The model's bias variation is minimized within the interval $[\lambda_{1se}, \lambda_{\min}]$ [26]. In this study, a λ value of 0.197 corresponded to a mean squared error (MSE) of 5.82, indicating the smallest error and the best fit for the model, with 8 significant factors identified. Consequently, we selected the top 8 variables ranked by feature importance in the Random Forest analysis: Self-Regulatory Fatigue, study time per day, time management disposition, absence of Interest in nursing-related fields, academic performance, Only Child, future professional prospects, and excessive course

difficulty. These variables were subsequently incorporated into the multiple linear regression analysis.

Multiple linear regression analysis of factors influencing academic burnout

Multiple linear regression analysis was conducted to evaluate whether the eight most significant factors identified from the LASSO regression model could predict levels of academic burnout. The regression analysis, with academic burnout as the dependent variable, revealed that six predictors: Self-Regulatory Fatigue, study time per day, time management disposition, academic performance, future professional prospects, and excessive course difficulty were significant factors influencing academic burnout among undergraduate nursing students ($P < 0.05$). These predictors explained 65.1% of the variance in academic burnout ($R^2 = 0.651$). Among the identified factors, Self-Regulatory Fatigue ($\beta = 0.449$) and excessive course difficulty ($\beta = 0.403$) were classified as risk factors, while time management disposition ($\beta = 0.392$) and positive future professional prospects ($\beta = 0.318$) acted as protective factors. Additionally, longer study time per day ($\beta = -0.184$) and higher academic performance ($\beta = -0.106$) were significantly associated with reduced levels of academic burnout. (Table 3).

Discussion

The aim of this study was to investigate the status of academic burnout among China undergraduate nursing students and identify its main influencing factors. The findings revealed that the mean score on the academic burnout scale for undergraduate nursing students was 55.29 ± 6.78 . Although in an uncertain intermediate state, it is approaching the critical threshold of 60 points and remains at a relatively high level. These results align with those reported by [7] and are slightly higher than the levels of academic burnout previously observed in general college student populations [27]. This difference may be attributed to the unique challenges faced by undergraduate nursing students, who are required to master both complex theoretical knowledge and practical clinical skills. We found that the prevalence of academic burnout among undergraduate nursing students in China was 57.5%, which is significantly higher than the rates reported in developed countries such as the United States and Italy [28]. This elevated rate may be attributed to the large population of undergraduate nursing students in China, as well as potential regional differences in educational systems and socioeconomic disparities. China's relatively low Human Development Index suggests that socioeconomic status is a significant risk factor for academic burnout, reflecting the availability of resources for educational expenses, such as tuition waivers and

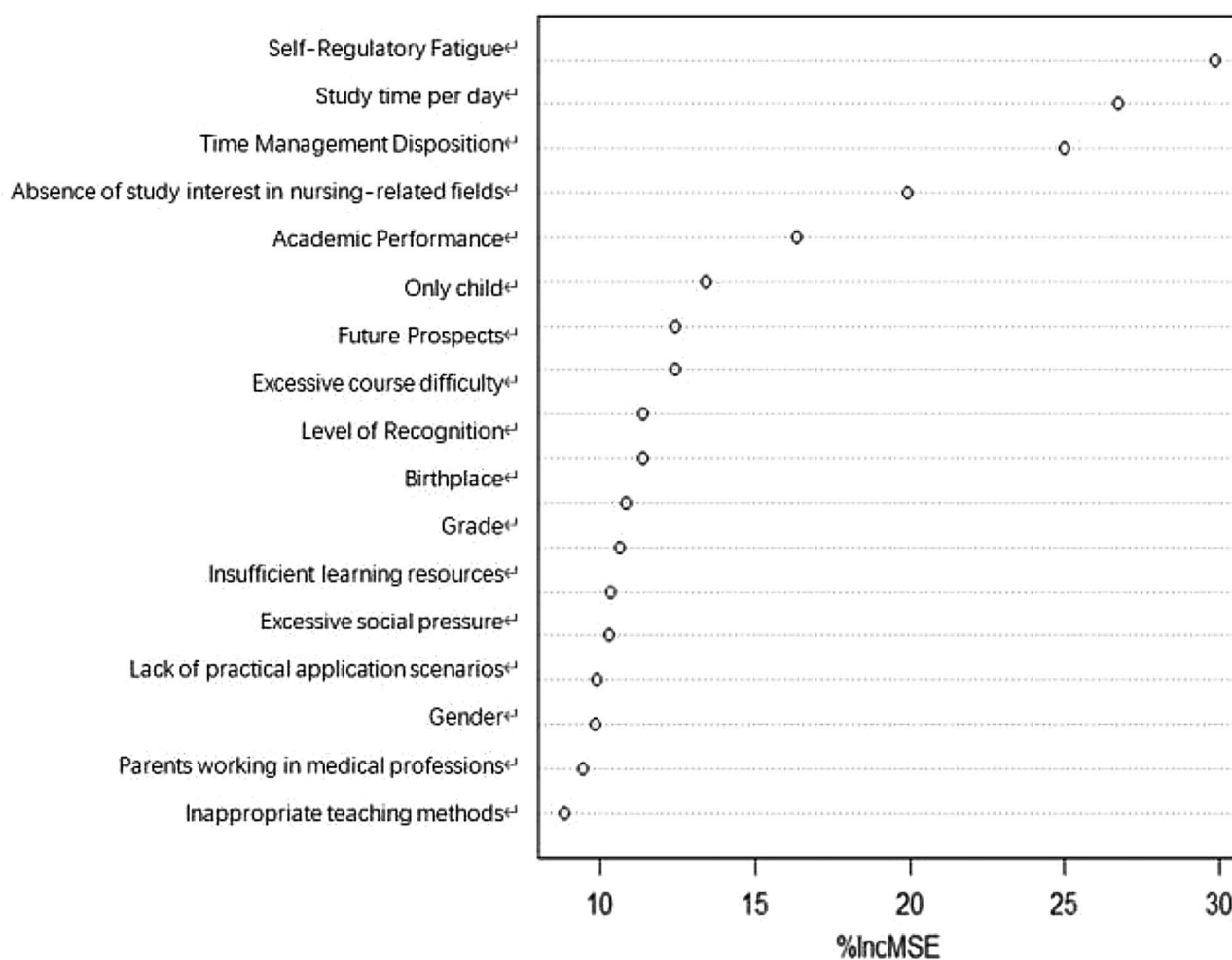
Table 2 Demographic characteristics and univariate analyses of the factors associated with ABS(N=442)

Variables	N(%)	ABS scores Mean ± SD	t/F	P
Gender			-5.36	<0.001
Male	104(23.52)	47.2 ± 18.11		
Female	338(76.48)	55.7 ± 15.13		
Grade			14.9	<0.001
Freshman	199(45.02)	59.4 ± 14.63		
Sophomore	90(20.36)	50.8 ± 16.91		
Junior	73(16.51)	55.3 ± 16.25		
Senior	80(18.09)	41.23 ± 16.48		
Birthplace			3.73	<0.001
Countryside	294(66.51)	57.52 ± 14.64		
Town	148(33.49)	50.89 ± 18.92		
Only child			-3.39	0.001
No	152(34.38)	51.41 ± 18.63		
Yes	290(65.62)	57.33 ± 14.86		
Student leader			-3.25	0.064
Yes	156(35.29)	53.20 ± 18.75		
No	286(64.71)	56.44 ± 15.00		
Study time per day			6.933	0.001
≤ 2 h	153(34.61)	57.28 ± 16.97		
3–5 h	183(41.40)	56.59 ± 14.96		
>6 h	106(23.99)	50.20 ± 17.35		
Barriers to learning				
Excessive course difficulty	229(51.80)	62.33 ± 10.45	-10.176	<0.001
Lack of practical application scenarios	195(44.11)	60.14 ± 13.07	-5.89	<0.001
Insufficient learning resources	135(30.54)	60.20 ± 14.55	-4.471	<0.001
Absence of study interest	118(26.69)	62.44 ± 13.78	-6.217	<0.001
Inappropriate teaching methods	98(22.17)	61.80 ± 13.33	-5.14	<0.001
Excessive social pressure	121(27.37)	60.24 ± 14.76	-4.169	<0.001
Family Issues Interference	45(10.18)	61.71 ± 14.73	-3.04	0.004
Academic performance			23.67	<0.001
≥ 4	32(7.23)	43.21 ± 20.05		
3-3.9	119(26.92)	57.03 ± 14.37		
2-2.9	172(38.91)	60.38 ± 12.26		
1-1.9	93(21.04)	53.74 ± 18.72		
0	26(5.88)	34.15 ± 12.05		
Parents working in medical professions			-3.83	<0.001
Yes	147(33.25)	50.85 ± 18.21		
No	295(66.75)	57.52 ± 15.09		
Main reasons for choosing nursing				
Personal preference	143(32.35)	56.44 ± 15.38	-1.043	0.298
Recommended by parents or others	208(47.05)	61.03 ± 13.10	-7.411	<0.001
Change of Major	80(18.09)	55.71 ± 17.20	-0.248	0.804
Good employment	194(43.89)	59.98 ± 12.99	-5.664	<0.001
Professional recognition			23.963	<0.001
Strongly agree	71(16.06)	50.45 ± 18.93		
Agree	180(40.72)	58.84 ± 13.28		
Neutral	138(31.22)	59.62 ± 14.60		
Disagree	28(6.33)	39.79 ± 15.52		
Strongly disagree	25(5.65)	37.08 ± 15.59		
Future prospects			29.447	<0.001
Very good	57(12.89)	48.36 ± 19.97		
Quite good	179(40.49)	59.01 ± 12.39		

Table 2 (continued)

Variables	N(%)	ABS scores Mean $\pm$ SD	t/F	P
Average	137(30.99)	60.38 $\pm$ 14.29		
Not very good	34(7.69)	48.26 $\pm$ 18.03		
Very poor	35(7.91)	34.60 $\pm$ 13.47		
SRF			135.222	<0.001
Low	207(46.83)	44.50 $\pm$ 17.04		
Medium	98(22.17)	63.64 $\pm$ 6.78		
High	137(30.99)	65.65 $\pm$ 8.52		
TMD			198.591	<0.001
Low	165(37.33)	40.60 $\pm$ 17.00		
Medium	47(10.63)	64.02 $\pm$ 6.30		
High	230(52.04)	64.06 $\pm$ 7.72		

Abbreviations: TMD, Time Management Disposition; SRF, Self-Regulatory Fatigue

**Fig. 1** Ranking of importance of factors influencing academic burnout

academic materials [29]. Consequently, the lack of such resources may serve as a considerable stressor contributing to the high prevalence of academic burnout among Chinese nursing students. Notably, within the three dimensions of academic burnout, the dimension of low

sense of accomplishment scored the highest. These findings underscore pervasive issues within the nursing student population related to unclear perceptions of the meaning of learning and low self-efficacy evaluations. These challenges may arise from external misconceptions

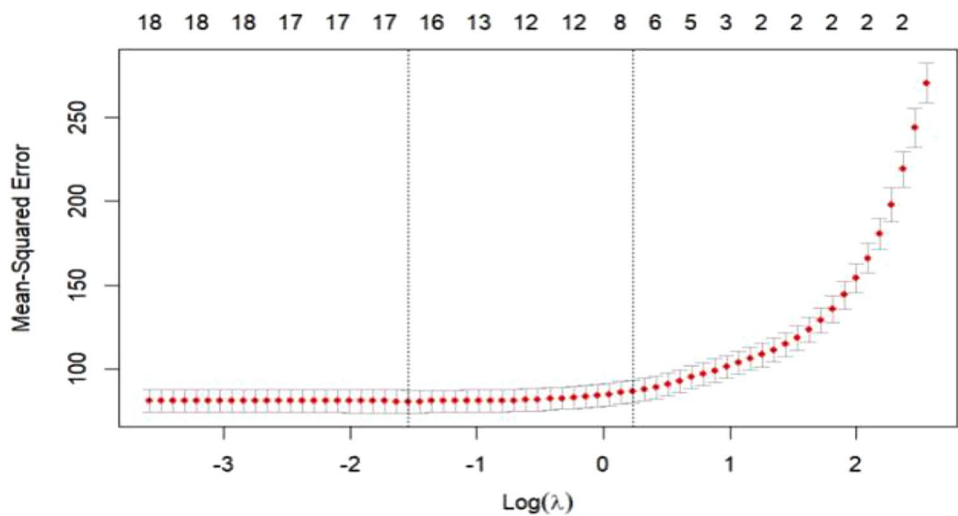


Fig. 2 Screening of feature variables based on LASSO analysis

Table 3 Multiple linear regression analysis of factors influencing academic burnout (N=442)

Predictor	B	SE	β	t	P
(intercept)	9.291	1.753		5.300	<0.001
TMD	−0.147	0.020	−0.392	−7.361	<0.001
SRF	0.611	0.072	0.449	8.435	<0.001
Academic Performance	−1.748	0.753	−0.106	−2.322	0.021
Excessive course difficulty(ref.NO)					
YES	13.263	1.353	0.403	9.805	<0.001
Absence of study interest(ref.NO)					
Yes	0.455	0.712	0.025	0.408	0.302
Only child(ref.NO)					
Yes	0.520	1.628	0.031	0.487	0.158
Study time per day(ref.<2 h)					
3–5 h	−0.685	1.781	−0.021	−0.385	0.701
>6 h	−7.083	2.055	−0.184	−3.447	0.001
Future Prospects(ref. Very good)					
Good	−10.655	2.234	−0.318	−4.768	<0.001
General	−12.029	2.316	−0.338	−5.194	<0.001
Bad	0.086	3.184	0.001	0.027	0.978
Very bad	13.751	0.155	0.226	4.538	<0.001

Note. F=412.814,P=0.000, adjusted R²=0.651.

and societal expectations that partially undermine academic beliefs, making it difficult for students to experience the sense of accomplishment they should ideally attain [30].Therefore, it is essential for Chinese nursing educators to focus on enhancing students’ sense of learning accomplishment, assisting them in establishing realistic achievement expectations, and providing more constructive learning resources to mitigate academic burnout among undergraduate nursing students.

The prevalence of academic burnout among undergraduate nursing students is increasing [31]. Our study established an influence factor analysis model based on Random Forest and LASSO regression to identify the significance of predictive factors and to screen for the

primary determinants of academic burnout. Furthermore, through multiple linear regression analysis, we ultimately identified the independent factors influencing academic burnout. A notable contribution of this study is the empirical support it provides for the Job Demands–Resources (JD-R) model. Our findings indicate that confronting job demands, such as Self-Regulatory Fatigue, excessive course difficulty, academic performance, and insufficient study time per day constitutes major risk factors for academic burnout among nursing students. Conversely, job resources such as effective time management disposition and positive future professional prospects emerge as strong protective factors, helping nursing students mitigate academic burnout.

Our research confirms that Self-Regulatory Fatigue is positively correlated with academic burnout, demonstrating a significantly greater influence than other variables. Additionally, study time per day, academic performance, and excessive course difficulty are identified as risk factors contributing to the occurrence of academic burnout. From a theoretical perspective, these findings validate the health impairment process of the Job Demands–Resources (JD-R) model within the context of undergraduate nursing students, as emphasized by [32].

Previous research [33, 34] has indicated that both self-control and cognitive tasks require individuals to activate and regulate controlled brain regions, leading to substantial energy expenditure and mental fatigue. This mental fatigue ultimately diminishes motivation to engage in tasks. Consequently, students experiencing high levels of Self-Regulatory Fatigue may struggle to cope with academic challenges, hinder their learning engagement, and develop negative attitudes, frustration, anxiety, and a diminished interest in learning, rendering them more vulnerable to academic burnout. Experimental research by [35] has demonstrated that being in a state of energy depletion extends the perceived time required for self-regulation, thereby increasing the likelihood of experiencing academic burnout. Therefore, to effectively alleviate academic burnout among undergraduate nursing students, nursing educators can incorporate Self-Regulatory training such as mindfulness into training programs. This approach enables the scientific management and protection of their limited psychological resources, preventing excessive depletion. The present study also confirmed that the prevalence of academic burnout was significantly higher among nursing students who perceived their coursework as excessively difficult. The overload of theoretical knowledge and complex skills training (emergency care simulation e.g.,) exceeded the students' current cognitive capacity, leading to a lack of "mastery experience." This chronic failure in developing competencies can trigger "competency attribution bias" [36], resulting in a vicious cycle of learned helplessness and academic burnout. Therefore, it is recommended that nursing educators systematically optimize the difficulty of undergraduate curricula based on cognitive load theory. For instance, in high-difficulty course segments, incorporating case-based guidance and providing timely positive feedback can help students develop a strong sense of self-efficacy. This approach effectively manages students' cognitive load and alleviates academic burnout.

The study revealed that academic burnout scores decreased with an increase in daily study time, a finding that aligns with the research conducted by [37]. Specifically, undergraduate nursing students who studied for more than six hours per day exhibited significantly lower levels of academic burnout. This suggests that nursing

educators should promote the incorporation of effective daily study time to help reduce levels of academic burnout among undergraduate nursing students. Academic performance is a critical factor that influences the future trajectory of nursing students and serves as an important indicator of educational quality and learning outcomes. An additional noteworthy finding from this study is that better academic performance among undergraduate nursing students is associated with lower levels of academic burnout. This observation corroborates the findings of [38] and suggests that academic performance may be a potential consequence of academic burnout. Research by [39] has demonstrated that academic burnout, characterized as a persistent and stable negative psychological state, has a more significant impact on achievement than academic mood. Furthermore, interpersonal stress and academic burnout have been shown to adversely affect academic achievement. Therefore, enhancing the academic performance of undergraduate nursing students could potentially mitigate their experiences of academic burnout.

According to the Job Demands–Resources (JD-R) model, time management disposition and positive future professional prospects serve as alternative work resources, acting as important protective factors in mitigating academic burnout among undergraduate nursing students. Our findings reveal that time management disposition have a negative predictive effect on academic burnout, indicating that undergraduate nursing students with stronger time management disposition experience lower levels of academic burnout. This is consistent with prior research [40] that underscores the beneficial role of time management in academic activities. Nursing students who excel in time management can effectively set goals and utilize their long-term objectives as motivation for their current academic tasks, thereby reducing the likelihood of developing academic burnout. Further research [41] shows that undergraduate nursing students with strong time management skills not only achieve significantly improved academic performance but also enhance their initiative and efficiency in learning, thereby mitigating the negative effects of academic burnout. To support this, we recommend the regular implementation of time management workshops designed to guide students in employing the SMART principle for goal setting. By teaching students how to allocate time effectively for each task and reduce energy loss associated with task switching, we can enhance their time management disposition and help prevent the onset of academic burnout.

Another key finding of this study was that undergraduate nursing students who maintained positive attitudes toward the future of the nursing profession experienced significantly lower rates of academic burnout compared to those with negative attitudes. A longitudinal study by

[42]. supported this finding, revealing that students with a pessimistic outlook on their careers scored an average of 37% higher in burnout levels. This phenomenon is closely linked to the escalating crisis of professional identity [43] amidst a global nursing shortage. Negative recognition of the nursing profession, undermine students' motivation to learn, while academic burnout exacerbates their alienation from the field, creating a vicious cycle. Therefore, it is imperative for nursing educators to integrate systematic professional identity courses into the core curriculum. By incorporating role-playing and simulation tasks based on clinical scenarios within these courses, we can enhance undergraduate nursing students' confidence in their career prospects and sense of agency. This approach has the potential to improve their perception of professional opportunities, thereby reducing the level of academic burnout.

Limitations

This study has some limitations. Firstly, the cross-sectional design employed in this study precludes the exploration of causal relationships between variables. Secondly, the sample was primarily selected using convenience sampling, with all participants originating from three medical schools and a high proportion of Freshman. This imbalance in institutional geographic distribution and grade levels may limit the generalizability and representativeness of the study findings. Finally, the reliance on self-reported measures may have compromised the reliability of the findings.

Conclusion

Our study indicated that academic burnout among undergraduate nursing students is at a moderately high level. The application of random forests combined with LASSO regression identified Self-Regulatory Fatigue, study time per day, time management disposition, academic performance, future professional prospect and excessive course difficulty as stable predictors of academic burnout, with Self-Regulatory Fatigue emerging as the most influential factor. These findings suggest that nursing educators should consider adopting a teaching framework centered on guided interaction to enhance nursing students' enthusiasm for exploration and problem-solving. By stimulating students' learning initiative and fostering a sense of accomplishment, educators can help balance cognitive resources and academic performance. Creating a positive learning atmosphere is essential for reducing academic burnout and promoting both academic development and professional growth among nursing students.

Abbreviations

ABS	Academic Burnout Scale
TMD	Time Management Disposition Scale
SRF-S	Self-Regulatory Fatigue Scale
DE	Dejection

IB	Improper Behavior
RPA	Reduced Personal Accomplishment

Supplementary Information

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Supplementary Material 1

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Author contributions

T. L: Conception and design, collection and assembly of data, data analysis and interpretation, manuscript writing. X.Q.S: data analysis and interpretation. J.L.W: Conception and design, article revision. M. Y, F.Y. L and Q. A: Provision of study materials, collection and assembly of data. J.Y.M and H.Y.X: Data analysis and interpretation. Collection and assembly of data. All authors have read and agreed to the published version of the manuscript.

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Data availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Declarations

Ethics approval and consent to participate

This study involved human participants and was approved by the Ethical Review Committee of Affiliated Hospital of Chengdu University of Traditional Chinese Medicine (NO.2023KL-098). All participants received an explanation of the purpose of the study and provided written informed consent before the investigation. Data were collected anonymously, respecting the confidentiality of the data collected in this study, in accordance with the Declaration of Helsinki. Participants were made aware that they could withdraw from the study at any time.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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